

Sunday 28<sup>th</sup> of June, 8:35 to 12:00



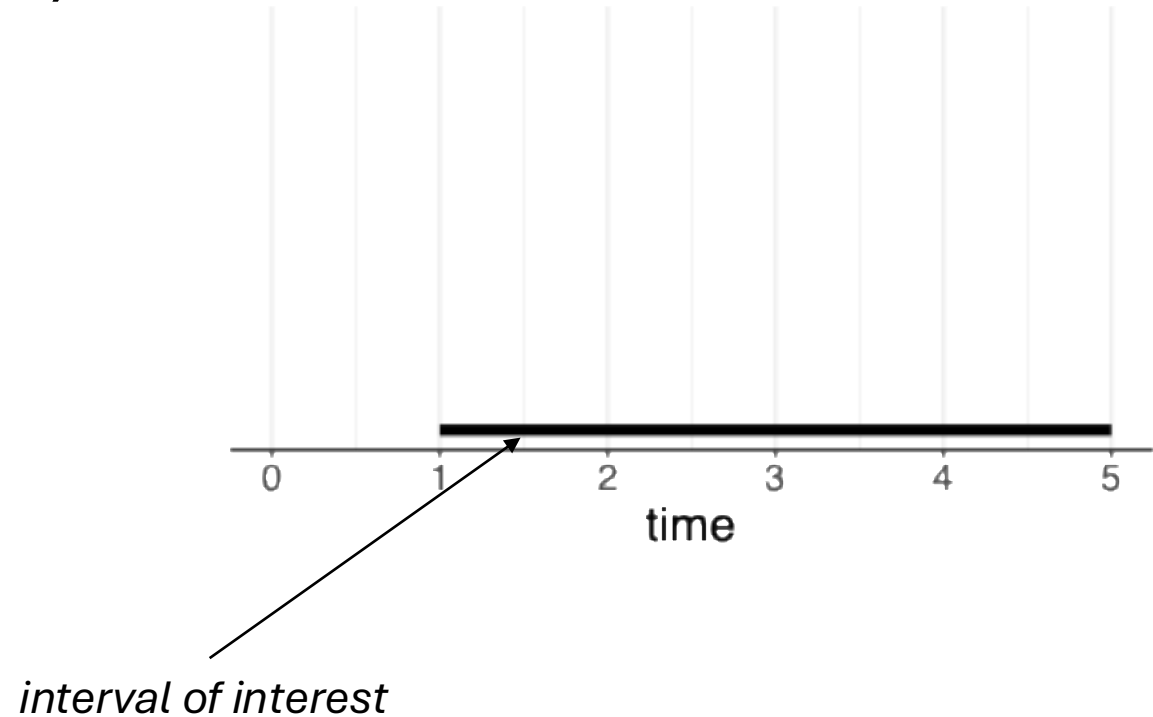
| Time     | Description                                                                                                                                                       | Discussant                              |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| [15 min] | (0) Introductions and administrivia                                                                                                                               | Trikalinos                              |
| [25 min] | (1) DES as a composition of point processes                                                                                                                       | Alarid-Escudero                         |
| [30 min] | (2) NHPPPs – key properties                                                                                                                                       | Trikalinos                              |
| [30 min] | (3) Sampling from NHPPPs                                                                                                                                          | Trikalinos                              |
| [15 min] | <b>Break</b>                                                                                                                                                      |                                         |
| [80 min] | (4) Working with code<br>-- Base R vs <code>nhppp</code> : Simulate first vs all events in $(a, b]$<br>-- Packaging a simple cancer DES<br>-- The vectorized case | [All]<br>Trikalinos<br>Sereda<br>Sereda |
| [5 min]  | (5) Advanced Topic Teaser on self-excitatory processes:<br>point processes that are not NHPPPs and when you may need them                                         | Trikalinos                              |
| [15 min] | General Q & A                                                                                                                                                     | All                                     |

# Section 2: Theory

The building block

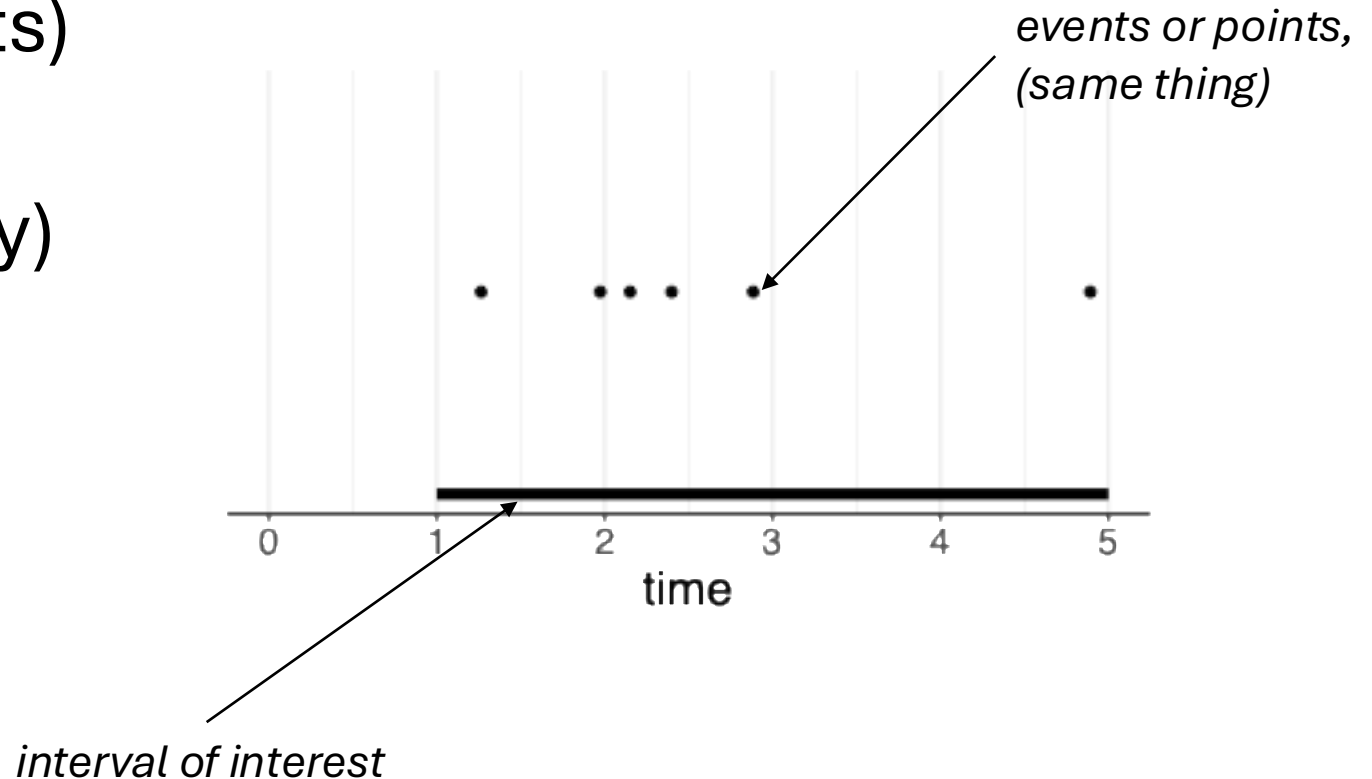
# The point process

- A scheme that generates a sequence of events (points) over a time interval



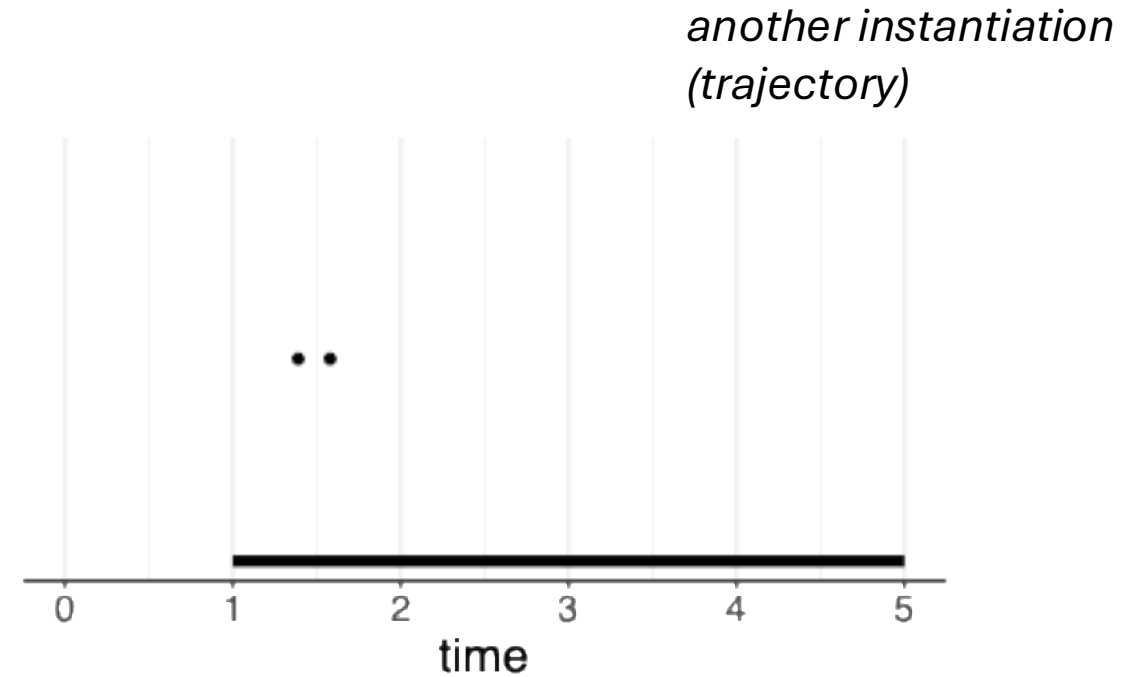
# The point process

- A scheme that generates a sequence of events (points) over time
- An instantiation (trajectory) of the process is a sequence of 0, 1 or more events in the interval, but none outside it



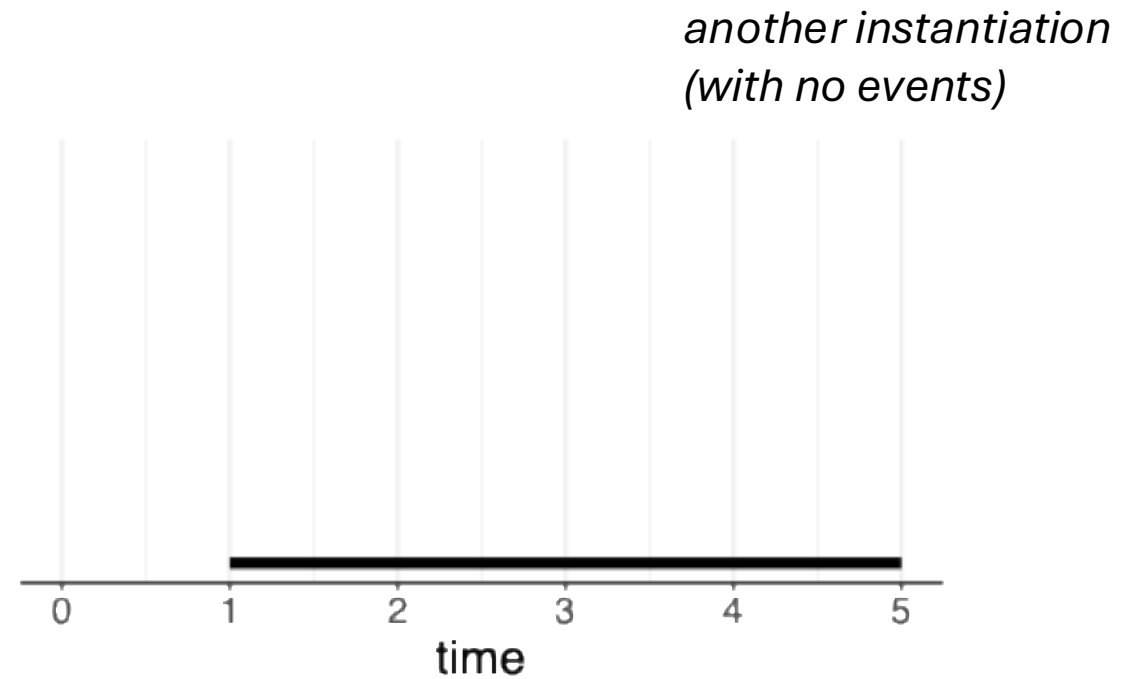
# The point process

- A scheme that generates a sequence of events (points) over time
- Each instantiation is random



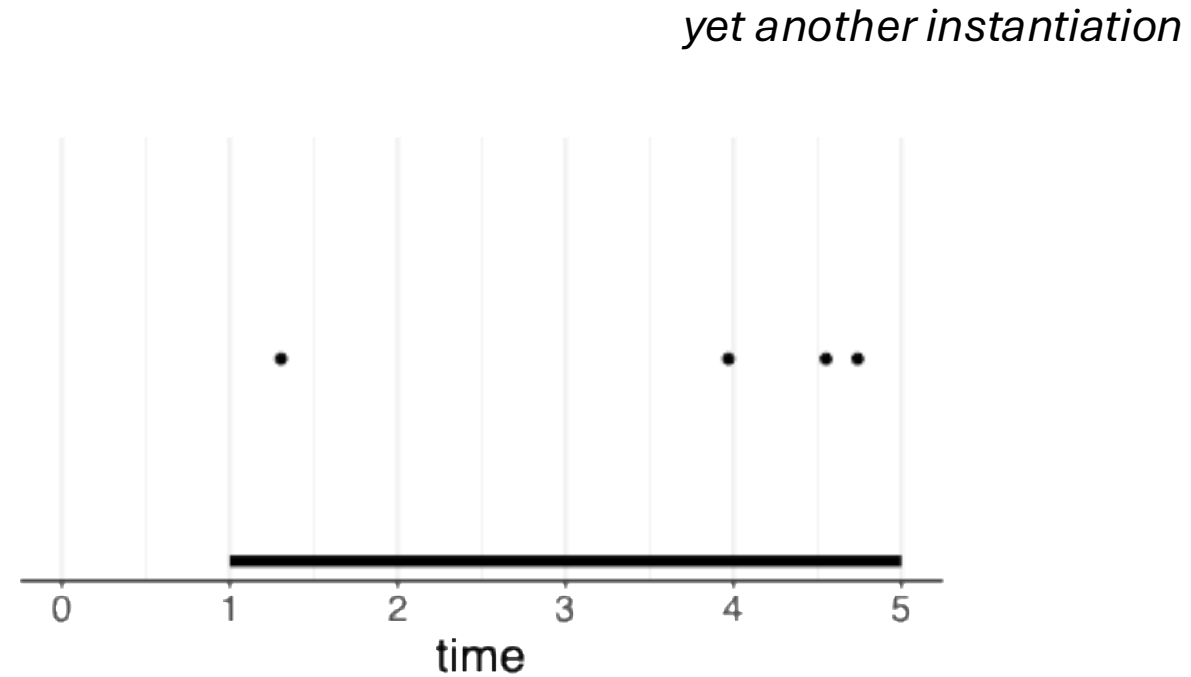
# The point process

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- Each instantiation is random



# The point process

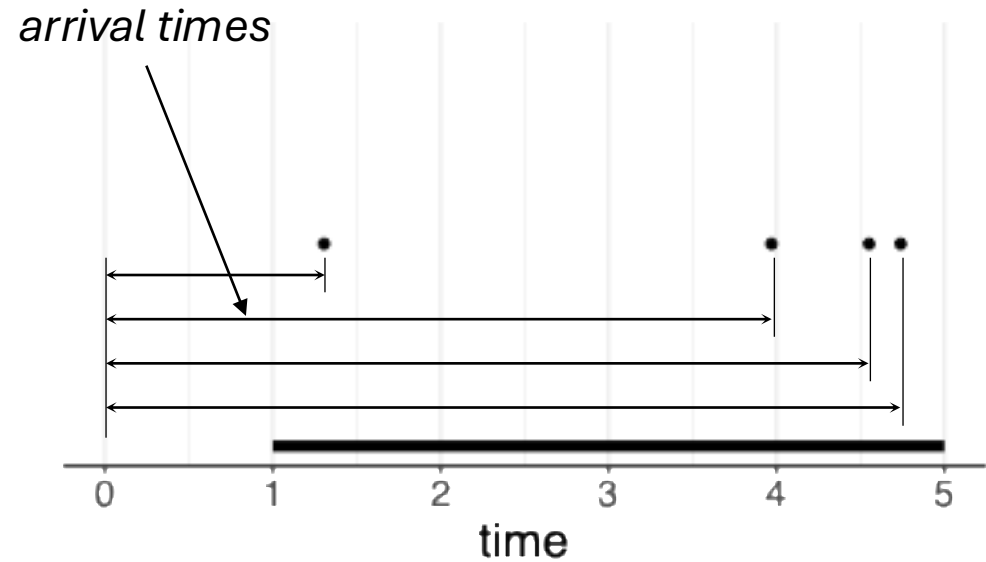
- A scheme that generates a sequence of events (points) over time
- Each instantiation is random





# The point process

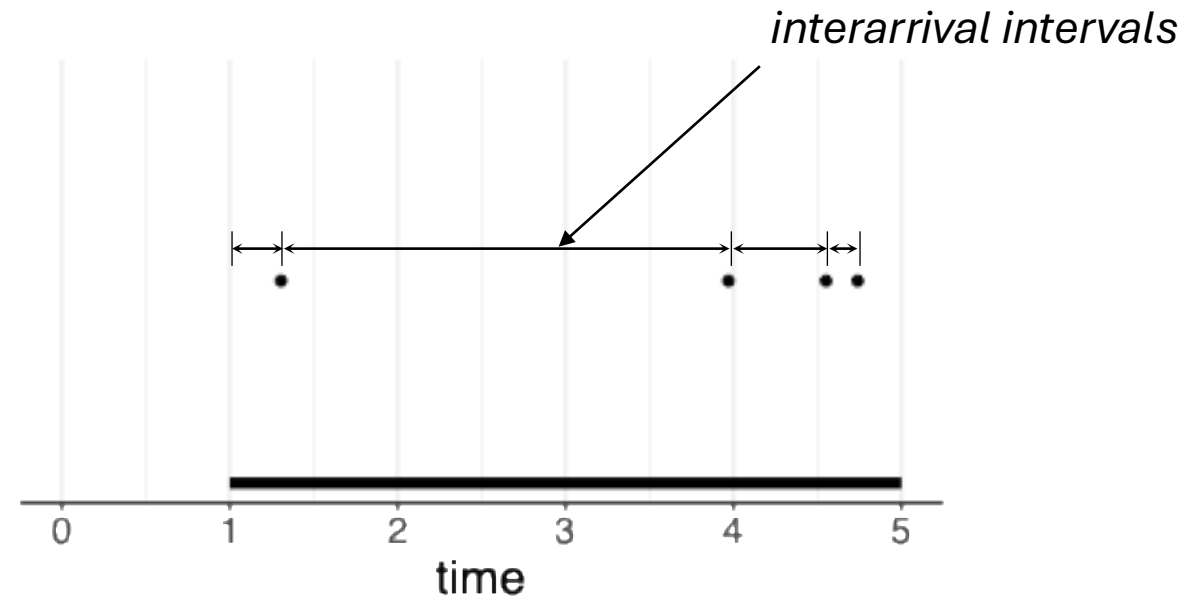
- The *arrival times* (times of the events) are random
- They start from whenever we zeroed the clock



# The point process

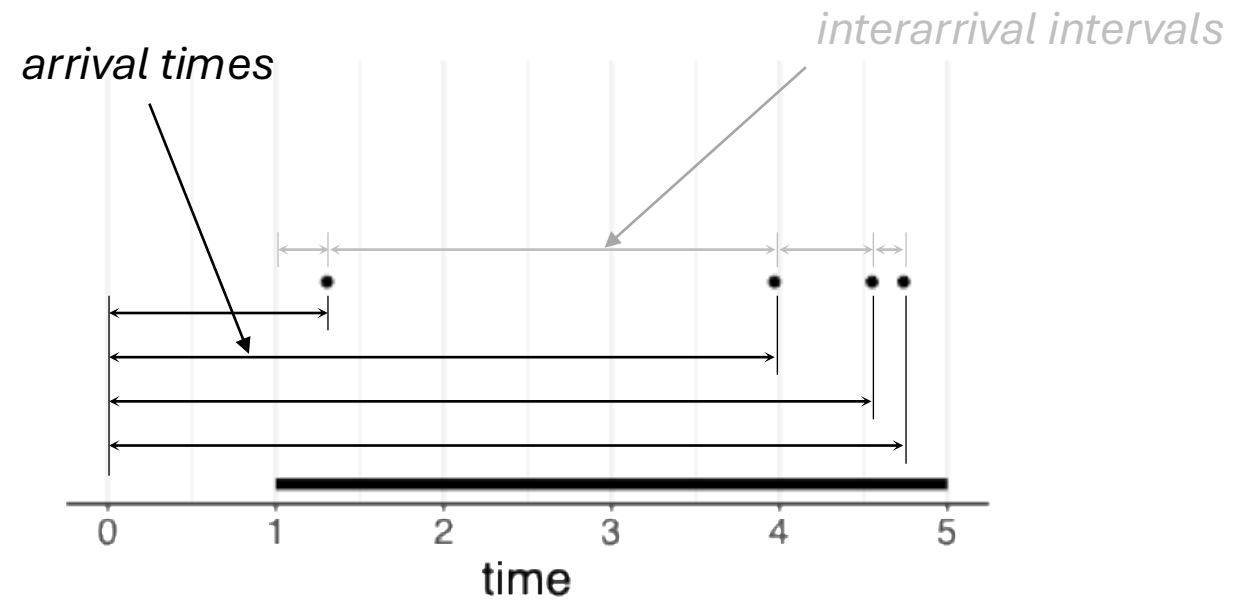
- The interarrival times are the lengths of the interarrival time intervals
- The arrival times and interarrival times give the same information

(... thus, the interarrival times are random)



# The point process

Hereon, we refer only to arrival times

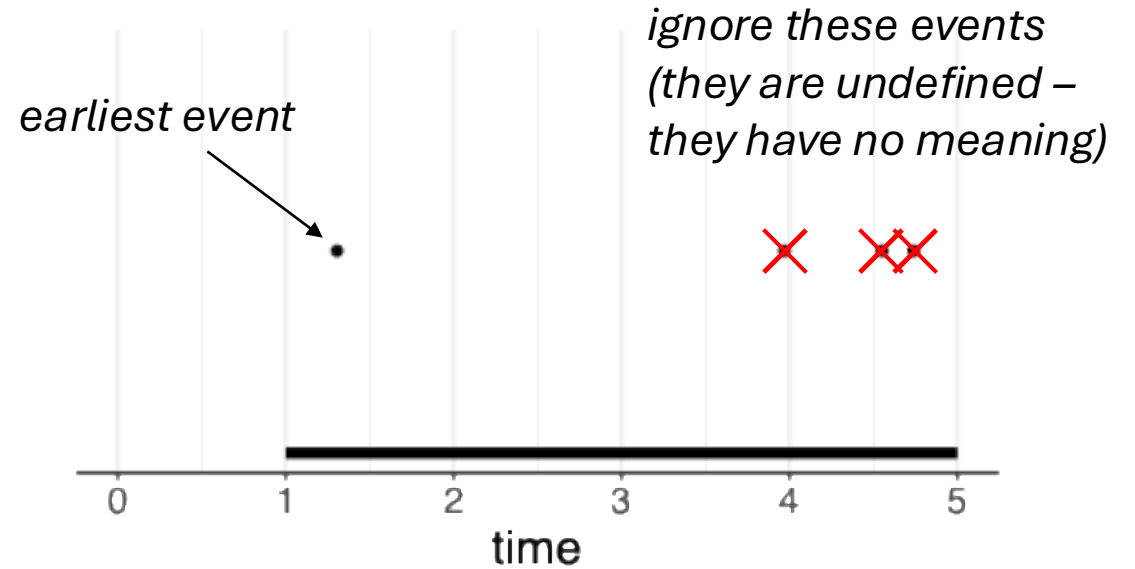


# Modeling non-repeatable events

If the point process models a *nonrepeatable* event, we care only about the **earliest event**.

Will it occur in the interval, and, and if so, when?

Example: model a cause of death

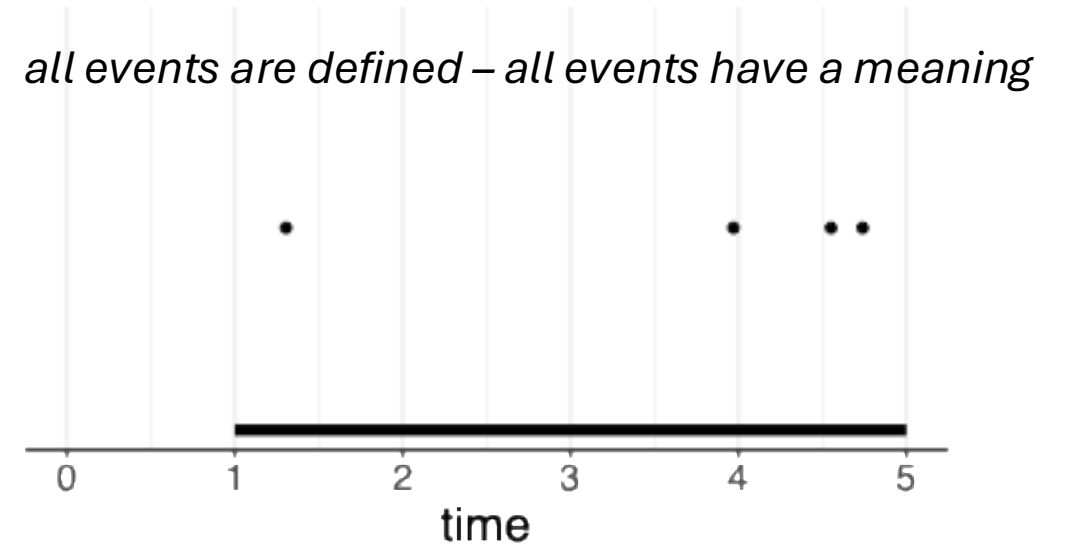


# Modeling repeatable events

If the point process models a *repeatable* event, we care are about **all events**.

Will any occur in the interval, and, and if so, when?

Example: model the emergence of tumors, or the start of symptomatic episodes



# The Poisson point process

- There are many types of point processes
- We will consider only a one type – the Poisson point process

# The Poisson point process

If for a sequence of events

$$\Pr[N(t, t + \Delta t) = 0] = 1 - \lambda \Delta t + o(\Delta t),$$

$$\Pr[N(t, t + \Delta t) = 1] = \lambda \Delta t + o(\Delta t),$$

$$\Pr[N(t, t + \Delta t) > 1] = o(\Delta t), \text{ and}$$

$$N(t, t + \Delta t) \perp\!\!\!\perp N(0, t),$$

*Number of events between  $t$  and  $t + \Delta t$*

*$o(\Delta t)$  becomes 0 **very fast***

for some  $\lambda > 0$  and as  $\Delta t \rightarrow 0$ ,  
then that sequence is a Poisson  
point process

# The Poisson point process (in English)

If for a sequence of events

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$$\Pr[N(t, t + \Delta t) = 1] = \lambda \Delta t$$

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*Over a vanishingly small interval*

- *you may get 1 event with probability  $\lambda \Delta t$  ...*



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- *you may get 1 event with probability  $\lambda \Delta t$  ...*
- *otherwise, you'll get 0 events;*

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- *you may get 1 event with probability  $\lambda \Delta t$  ...*
- *otherwise, you'll get 0 events;*
- *you'll never get many concurrent events*

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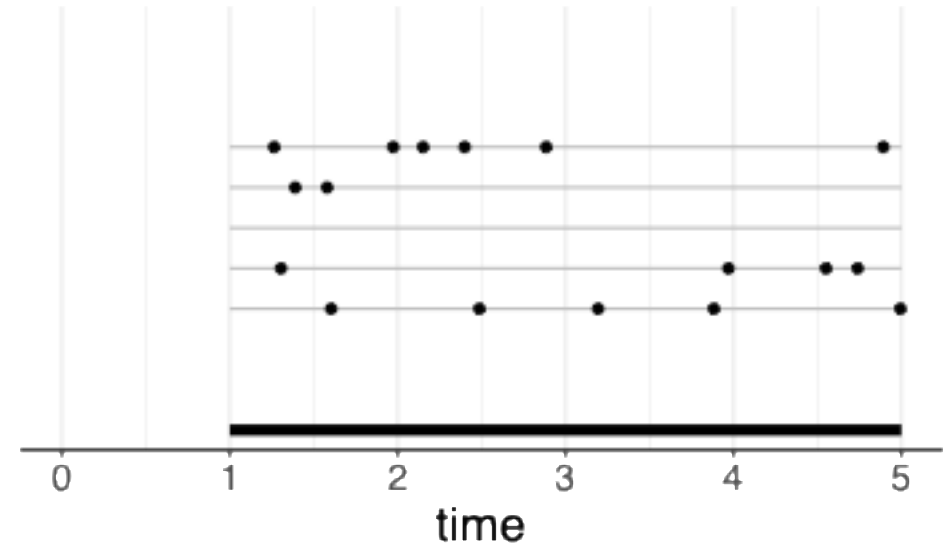
for some  $\lambda > 0$  and as  $\Delta t \rightarrow 0$ ,  
then that sequence is a Poisson  
point process

*Over a vanishingly small interval*

- *you may get 1 event with probability  $\lambda \Delta t$  ...*
- *otherwise, you'll get 0 events;*
- *you'll never get many concurrent events*
- *and it does not matter what happened in the past*

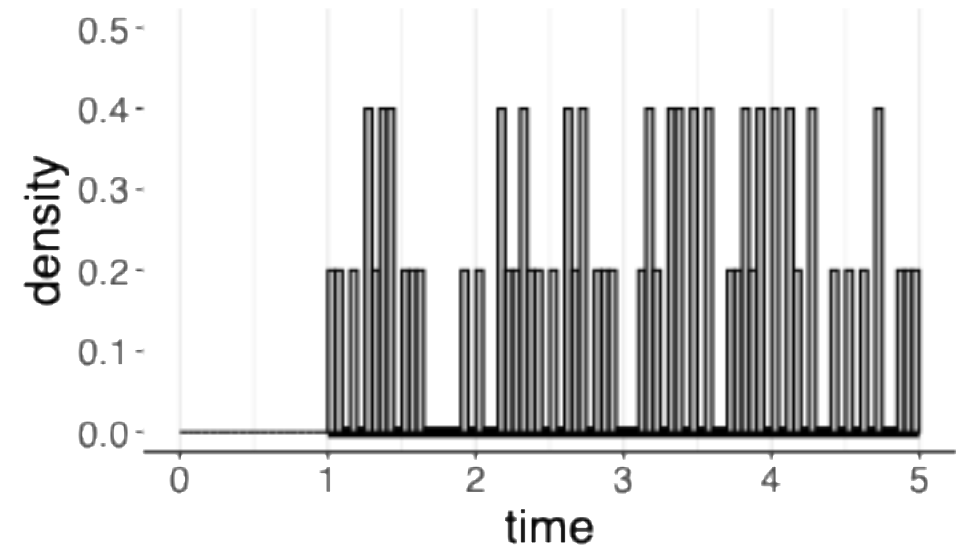
# The intensity function $\lambda$ in the example

Event times for five  
instantiations



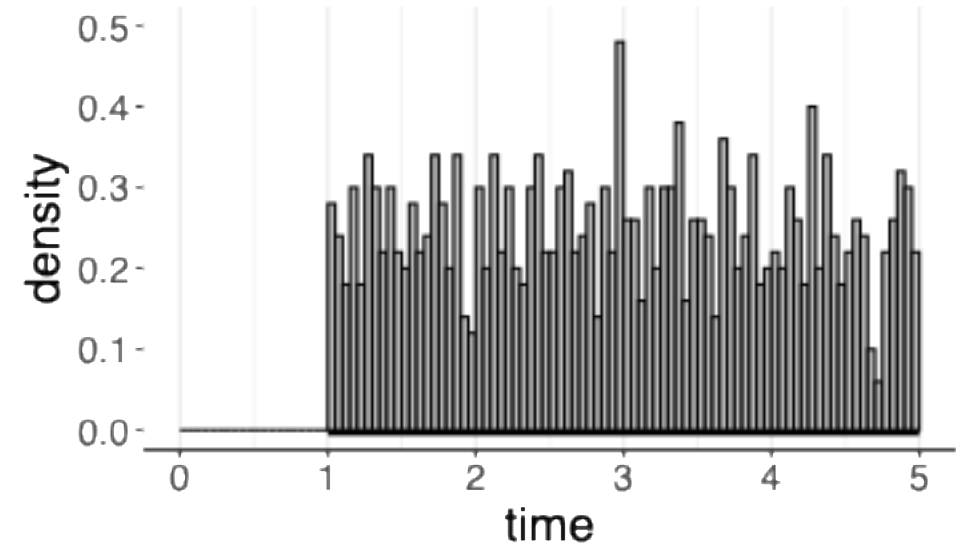
# The intensity function $\lambda$ in the example

A histogram of *pooled* event times for 100 instantiations



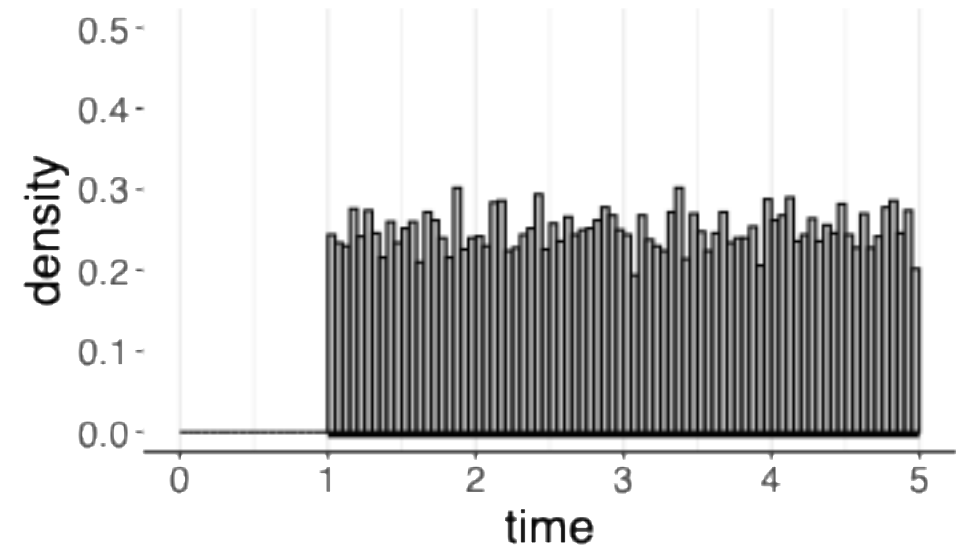
# The intensity function $\lambda$ in the example

... for 1000 instantiations



# The intensity function $\lambda$ in the example

... and for 10000 instantiations

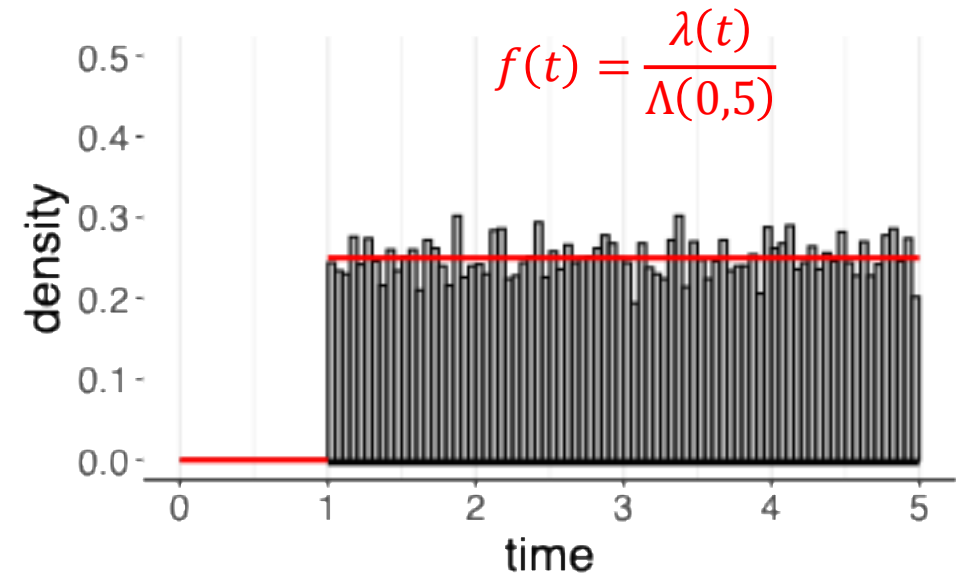


# The intensity function $\lambda$ in the example

As the number of instantiations goes to infinity, the histogram approaches the shape of the intensity function  $\lambda(t)$ .

The intensity function governs event occurrence.

(It is the same object as the hazard rate function in survival analysis)



$f(t)$ : The intensity function scaled by the expected number of events in the interval



# Time-homogeneous and non-homogeneous

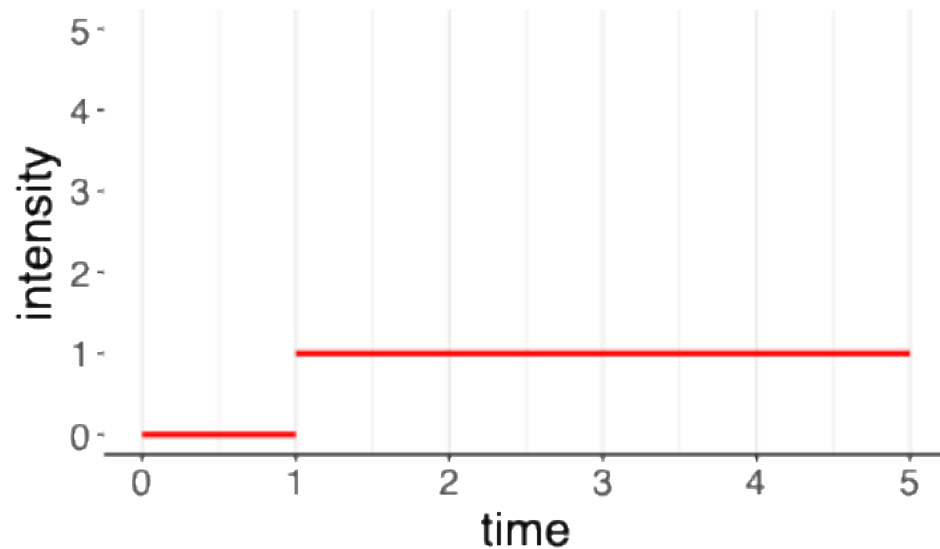
- $\lambda(t) = \text{constant}$ : the Poisson point process (PPP) is called time-homogeneous
- Otherwise, it is called a non-homogeneous PPP (NHPPP)

# The three important objects

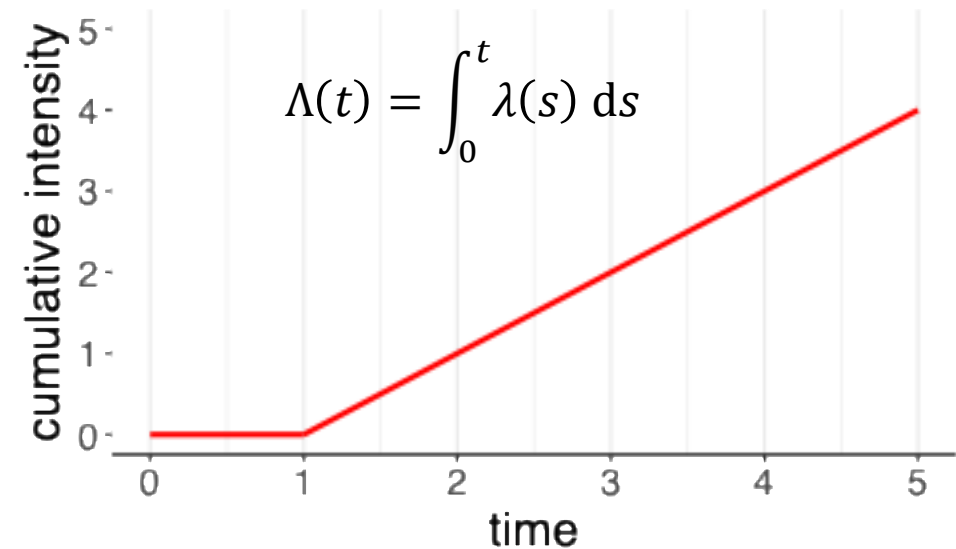
- Intensity function  $\lambda(t)$ 
  - *Always available*
  - *Sufficient to sample from any NHPPP efficiently and accurately*
- Cumulative intensity function
$$\Lambda(t) = \int_0^t \lambda(s) \, ds$$
- Inverse cumulative intensity function  $\Lambda^{-1}(z)$ , defined so that  $\Lambda^{-1}(\Lambda(t)) = t$ 
  - *Not always available*
  - *If available, you accelerate sampling by several times*

# Intensity and cumulative intensity functions

Intensity function  $\lambda(t)$

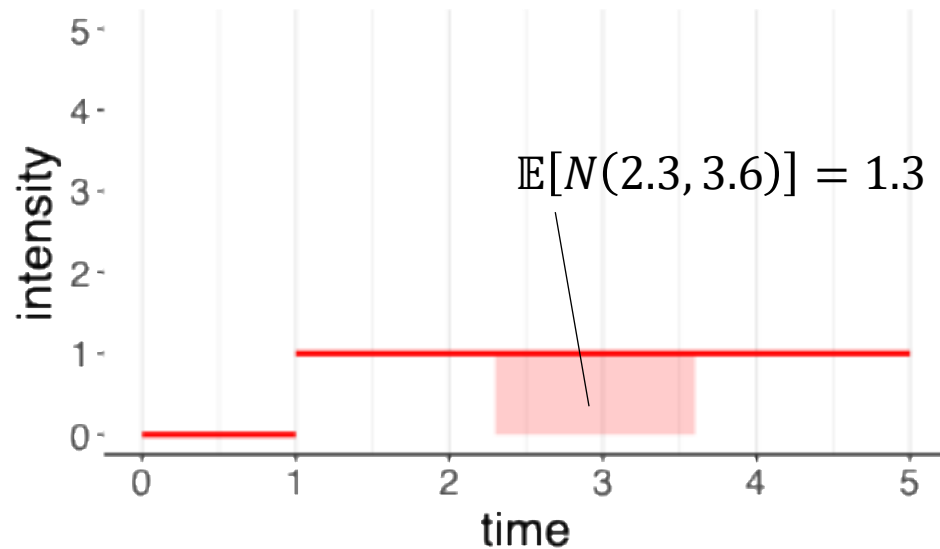


Cumulative intensity function  $\Lambda(t)$

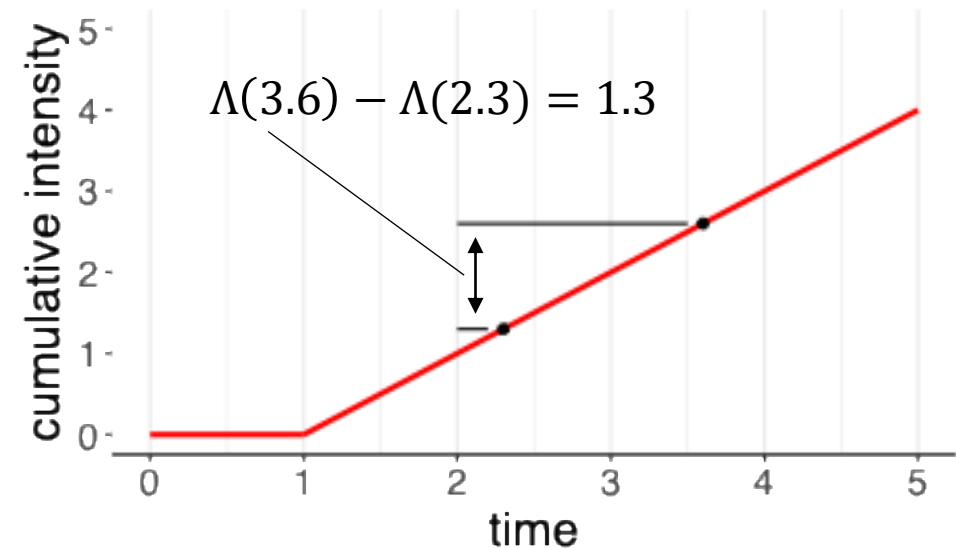


# Intensity and cumulative intensity functions

Intensity function  $\lambda(t)$

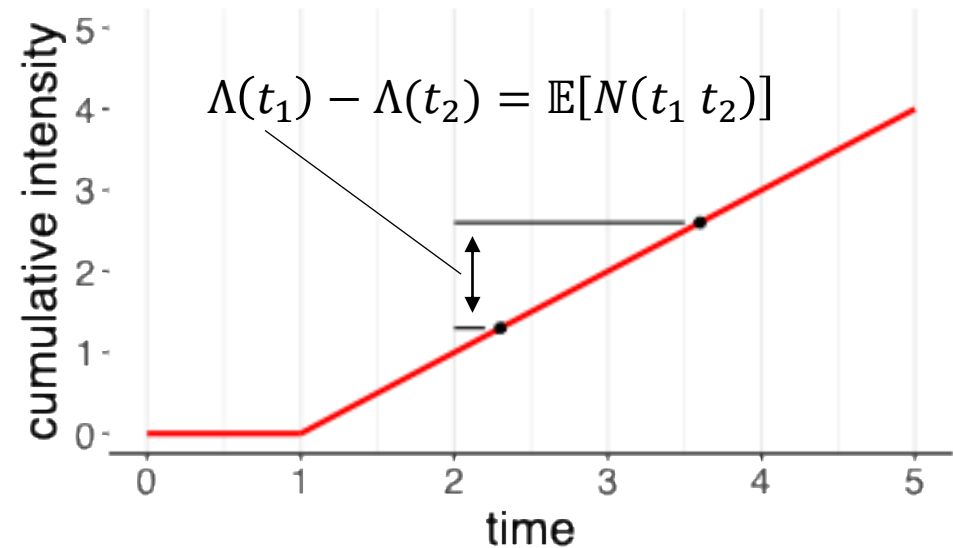
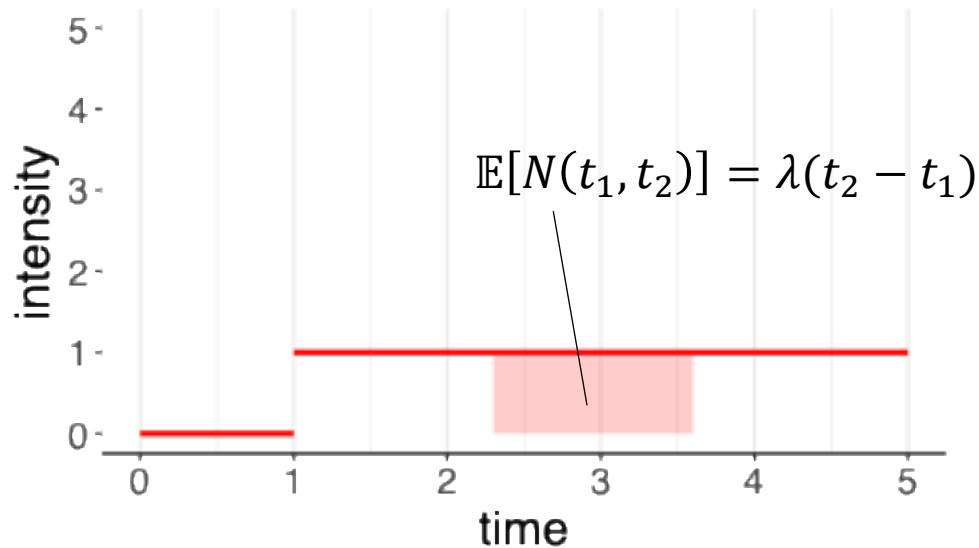


Cumulative intensity function  $\Lambda(t)$



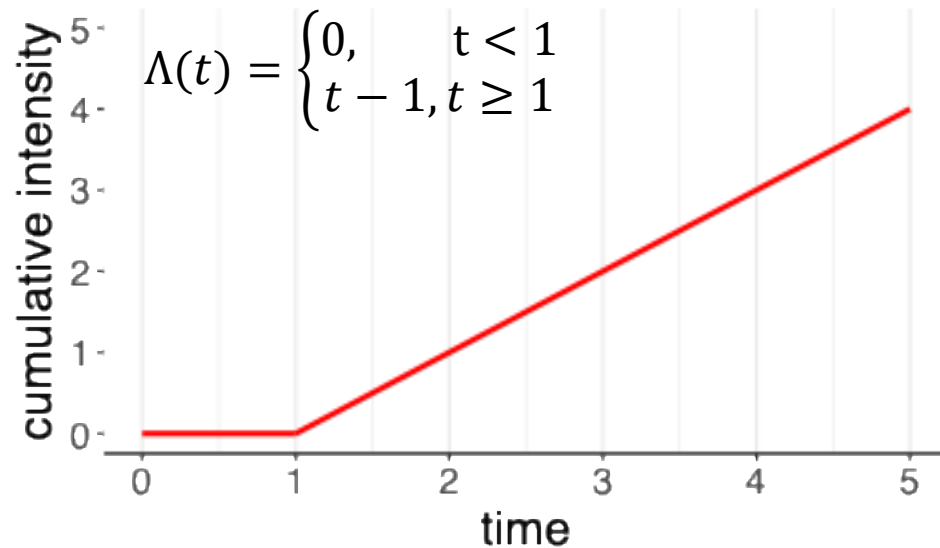
# Intensity and cumulative intensity functions

$N(t_1, t_2) \sim \text{Poisson}(\Lambda(t_2) - \Lambda(t_1))$ ,  
irrespective of the form of  $\lambda(t)$

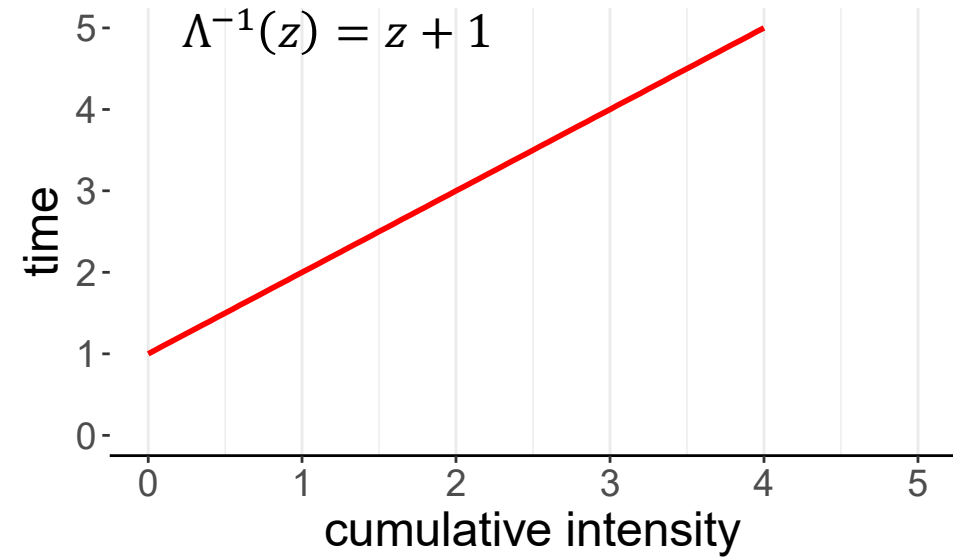


# Cumulative intensity function and its inverse

**Cumulative intensity function  $\Lambda(t)$**



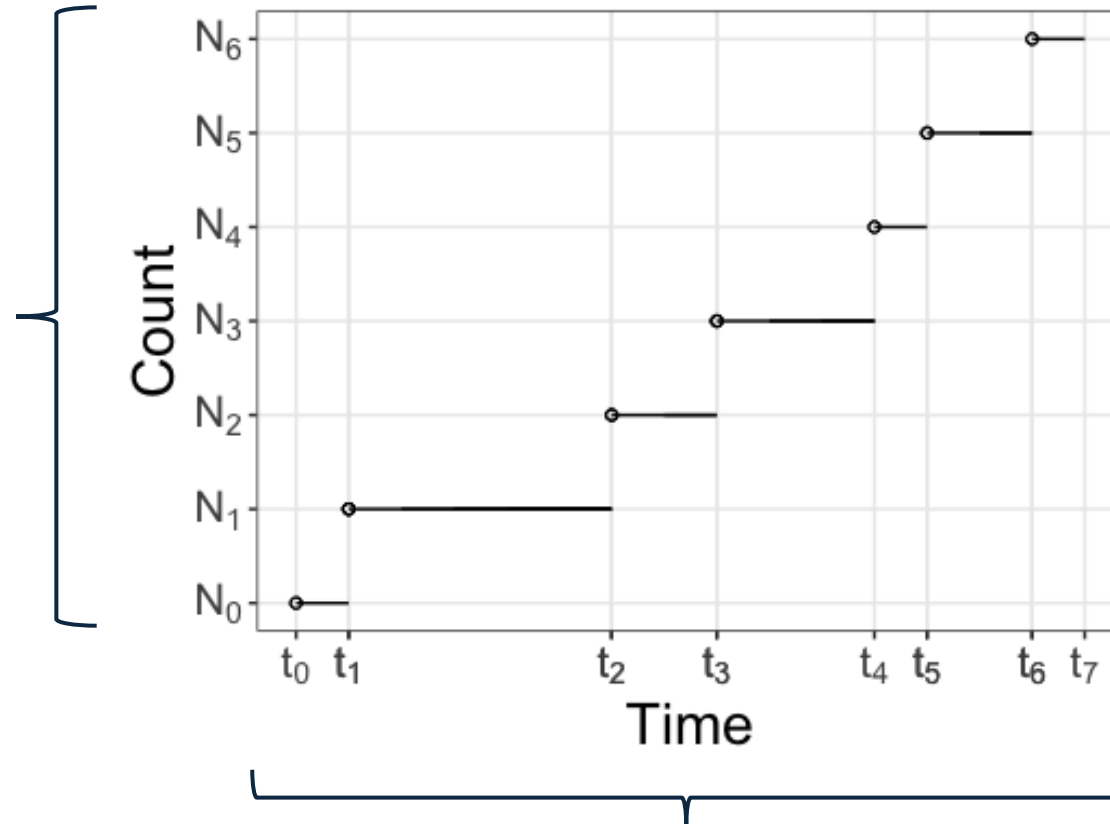
**Inverse cumulative intensity function  $\Lambda^{-1}(z)$**



# Duality with the Poisson counting process

Poisson *counting*  
*process*

$N_0, N_1, \dots$   
Cumulative number  
events over time



Poisson *point process*

$t_0, t_1, \dots$

# How does this connect to survival analysis?

- The theory of point processes is the foundation of survival analyses
- Often survival analysis is about *the time to the earliest event*
- The intensity function is the same as the hazard rate function
- The cumulative intensity function is the same as the (integrated or cumulative) hazard function
- All the tools described here can be used for statistical simulations for survival analysis



# Distribution of times for earliest vs (a pooled) event

Survival function of first-event:

$$\Pr[Z_{(1)} > t] = \Pr[N(a, t) = 0] = e^{-\Lambda(a, t)}$$

Density of times in  $(a, b]$  of pooled events (from all instantiations)

Density of

- time to first-event

$$g(t) = \lambda(t)e^{-\Lambda(a, t)}$$

$$f(t) = \frac{\lambda(t)}{\Lambda(a, b)}$$

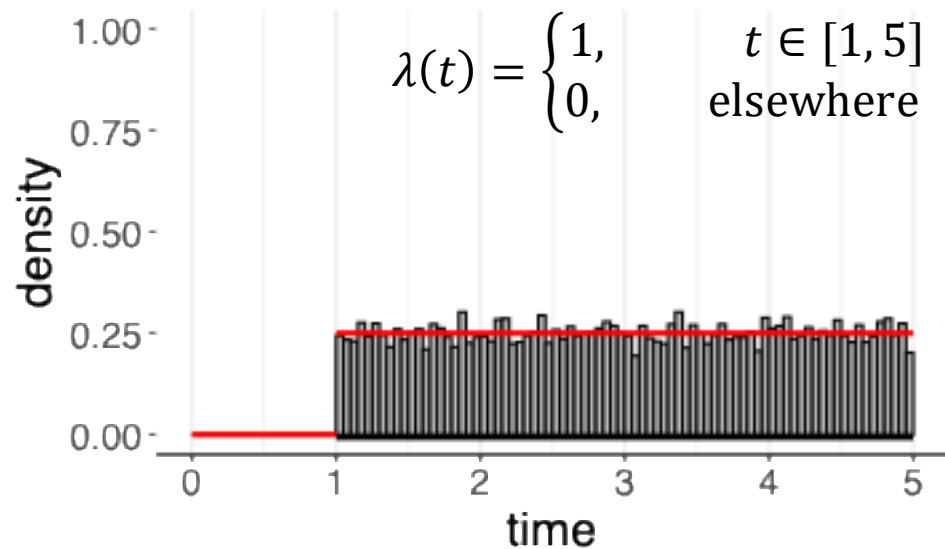
- time to first-event in  $(a, b]$

$$g_{(a, b]}(t) = \frac{\lambda(t)e^{-\Lambda(a, t)}}{1 - e^{-\Lambda(a, b)}}$$

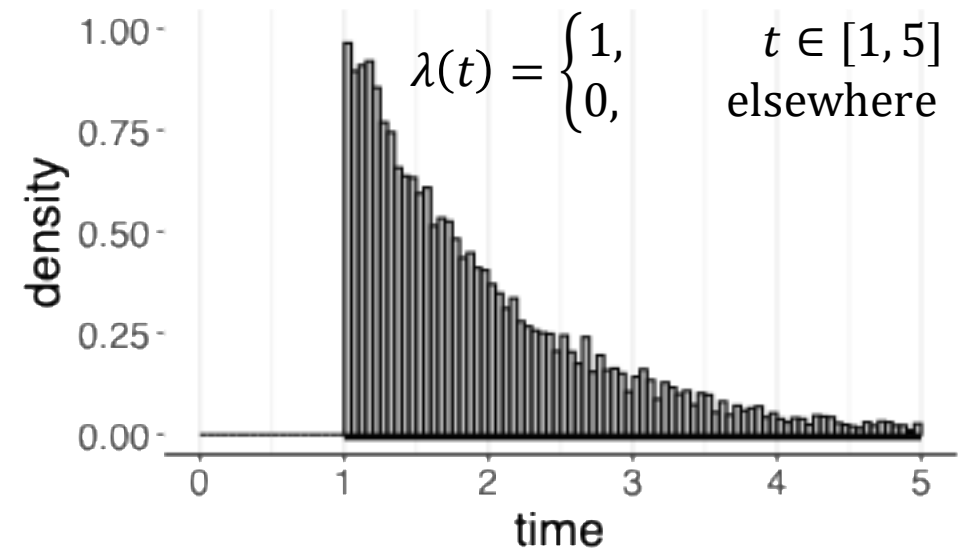
*(i.e., distributed according to the normalized intensity measure)*

# All events vs earliest event in the example

All events, 10K instantiations



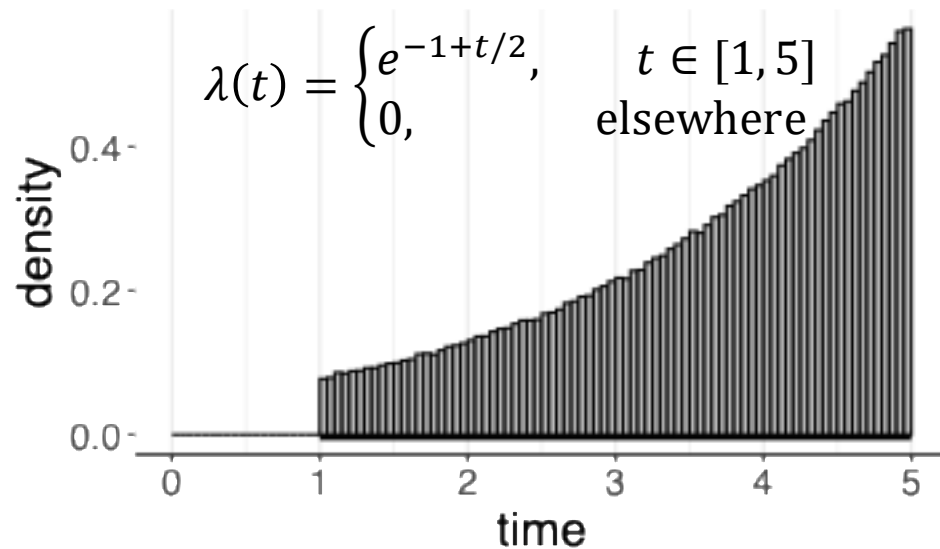
Earliest event, 10K instantiations



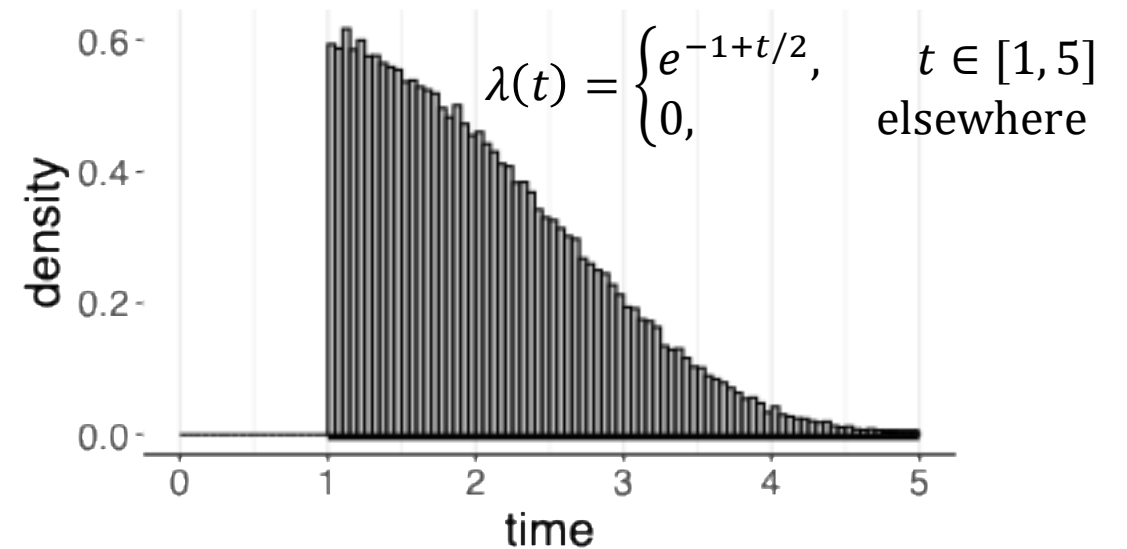
*The histogram of the earliest event times does not approach the shape of the intensity function*

# All events vs earliest event, different example

All events, 100K instantiations



Earliest event, 100K instantiations



Next ... Section 3: Sampling

Sunday 28<sup>th</sup> of June, 8:35 to 12:00



| Time     | Description                                                                                                                                                       | Discussant                              |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| [15 min] | (0) Introductions and administrivia                                                                                                                               | Trikalinos                              |
| [25 min] | (1) DES as a composition of point processes                                                                                                                       | Alarid-Escudero                         |
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| [30 min] | (3) Sampling from NHPPPs                                                                                                                                          | Trikalinos                              |
| [15 min] | <b>Break</b>                                                                                                                                                      |                                         |
| [80 min] | (4) Working with code<br>-- Base R vs <code>nhppp</code> : Simulate first vs all events in $(a, b]$<br>-- Packaging a simple cancer DES<br>-- The vectorized case | [All]<br>Trikalinos<br>Sereda<br>Sereda |
| [5 min]  | (5) Advanced Topic Teaser on self-excitatory processes:<br>point processes that are not NHPPPs and when you may need them                                         | Trikalinos                              |
| [15 min] | General Q & A                                                                                                                                                     | All                                     |